

The Solution to the Malfatti Problem in Sanpo Tenzan Tebikigusa

A Comparison between Kazuhide Omura's Method
and a Maxima-based Approach

Setsuo Takato

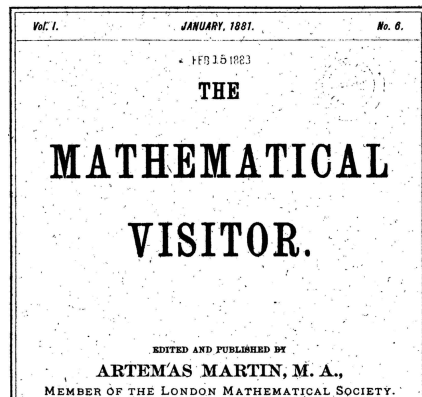
KeTCindy Center • Hon. Member of JSME

2026.09.12

Malfatti Circles

Brief History

- (1) Malfatti 1803
- (2) Ajima, Fukyu Sanpo
before 1773
- (3) Fujita, Seiyo Sanpo 1781
- (4) Omura
Sanpo Tenzan Tebikigusa
1841
- (5) Myojorinji 1865
- (6) The Mathematical Visitor, 1, 1881
Edmunds, Knisely, Seitz
- (7) Solution using Maxima and mnrc method 2025-



Solution in Seiyo Sanpo (Fujita)

- (1) Multiply middle and small diameters, take root, name it Ten

$$Ten = \sqrt{2r_2 \times 2r_3} = 2\sqrt{r_2 r_3}$$

- (2) Add all 3 diameters, multiply by large, take root, name it Chi

$$Chi = \sqrt{(2r_1 + 2r_2 + 2r_3) \times 2r_1} = 2\sqrt{r_1(r_1 + r_2 + r_3)}$$

- (3) Multiply large by Ten, divide by (Chi - large) to get full diameter

$$2r = \frac{2r_1 \times Ten}{Chi - 2r_1}$$

- (4) Substitute (1) and (2) into Ten and Chi of (3) and simplify

$$2r = \frac{2r_1 \times 2\sqrt{r_2 r_3}}{2\sqrt{r_1(r_1 + r_2 + r_3)} - 2r_1} = \frac{2r_1 \sqrt{r_2 r_3}}{\sqrt{r_1(r_1 + r_2 + r_3)} - r_1}$$

Derivation in SanpoTenzanTebikigusa

D_l, D_m, D_s : Diameters of 3 circles

(1) Definition of variables

$$Ne = \sqrt{D_l \cdot D_s}, Ushi = \sqrt{D_m \cdot D_s}$$

$$Ne + Ushi = U(, Tatsu, Mi)$$

(2) Construction of relations (Equation setup)

Write direct product on left

Eliminate direct product/solve”

(3) Elimination and derivation of algorithm

(4) Multiply middle and small diameters,take root(Rin)

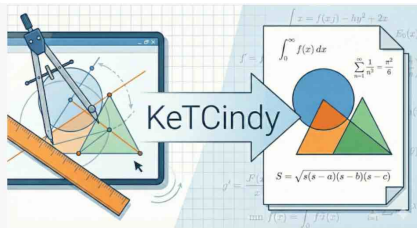
$$Rin = \sqrt{D_m \cdot D_s}, Ho = \sqrt{(D_l + D_m + D_s) \cdot D_l}$$

$$D = \frac{Rin \cdot D_l}{Ho - D_l}$$

Solution by KeTCindy

What is KeTCindy

- Macro package for D
- Interactively create T
 - (1) Create source co
 - (2) Execute T_EX via
 - (3) Call Maxima (C
- Distribution of KeTC
 - Search for "ketcin
 - KeTTeX ($\subset$ TeXLive
 - Students from Nu
 - created a batch in



KeTCindy Home

KeTCindy Center
2026-08-19

最新ニュース

- 2026-08-19 KeTCindy が更新されました (version 4.5.56)
 - ketslideが%ualatex対応になりました。
 - KeTCindyのインストールを参考にしてください。
 - 必須環境: TeX (KeTTeX, TeXLive) と R
 - Windowsの場合: Sumatra PDFも必要です。
 - **重要:** インストール実行時に「すべてのユーザに対してインストール」を選択してください。 (インストール先が C:\Program Files\SumatraPDF になります)
- 2026-06-12 Cinderella が更新されました (build2136)
 - Cinderellaのインストールを参考にしてください。
 - 検索時は beta cinderella としてください。
- 2026-05-19 KeTTeX が TeXLive20260313 に更新されました
 - 「kettex」または「kettex release」で検索可能です。
 - KeTTeXのインストール / エディタ設定(TeXWorks等)
- Windows用 一括インストーラ (沼津高専 学生作成)
 - Cinderella, R, Sumatra, KeTTeX, KeTCindy が一括でインストールされます。

Wasan and Solution by Maxima

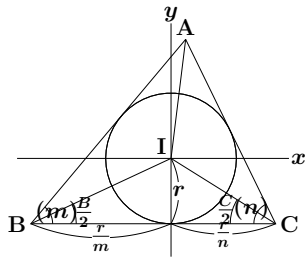
- Wasan geometry are interesting teaching materials. Solving them hands-on will increase interest and engagement
- Solving directly requires insight and considerable calculation skills
- Using CAS, high school level geometry knowledge is sufficient
- Even with CAS, it is often not easy to solve
 - Calculations involving radicals usually fail
- Thus, we devised the mn timer method to express triangles using $m = \tan \frac{B}{2}$, $n = \tan \frac{C}{2}$, and in-radius r .

The mnR Method

- $m = \tan \frac{B}{2}, n = \tan \frac{C}{2}$

$r = \text{inradius}$

$$\angle B = (m), \angle C = (n)$$



- Expressions of triangle properties (Rational expressions)

$$\text{vtxL}:(-\frac{r}{m}, -r), \text{vtxR}:(\frac{r}{n}, -r), \text{vtxT}:(\frac{r(n-m)}{1-mn}, \frac{1+mn}{1-mn})$$

$$\text{edgB}:\frac{r}{m} + \frac{r}{n}, \text{edgL}=\frac{r(1+m^2)}{m(1-mn)}, \text{edgR}:\frac{r(1+n^2)}{n(1-mn)}$$

Incircle inC, inR, Circumcircle cirC, cirR

Excircles exCa, exRa, exCb, exRb, exCc, exRc

- Created Maxima library for the mnR method

Main Commands of mnr Method

- Basic commands

putT(m,n,r) Place triangle with incenter at origin

slideT(p1,p2) Translate so that p1 matches p2

rotateT(m,p) Rotate by $2 \tan^{-1}(m)$ around p

- Main commands for formula manipulation

numerf(f),denomf(f) Factorize num and denom

frfactor(f) Simplify fraction

frev(eq,rep) Substitute rep into eq and simplify

radcan(s) Simplify irrational expressions(Maxima)

Main Commands for Geometry

- Angle conversions

$\text{supA}(m), \text{comA}(m)$ Supple./comple. angles

$\text{plusA}(m1, m2), \text{minusA}(m1, m2)$ Sum/diff. of angles

$\text{timesA}(nn, m)$ Integer multiples of angles

- Others

$\text{dotProd}(v1, v2), \text{crossProd}(v1, v2)$ Inner/outer prod

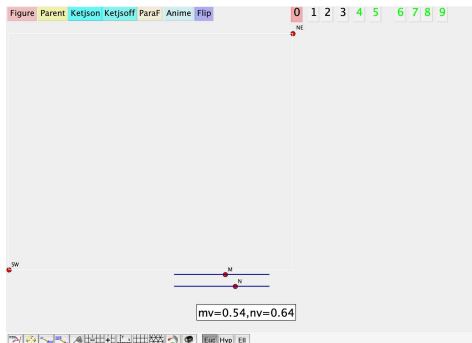
$\text{lenSeg2}(p1, p2)$ Square of distance

$\text{meetLine}(pts1, pts2)$ Intersection point

$\text{comTan1}, \text{comTan1C}$ Common outer/inner tangent

Solution by KeTCindy 1

- (1) Prepare Cinderella (cdy) file and mkcmd file
e.g., malfattiin.cdy, malfattiinmkcmd.txt
- (2) Use template file for cdy (import mkcmd.txt)



Screen of malfattiin.cdy

```

1 Ketinit();
2 Setwindow([-7,5],[-5,5]); Addax(0);
3 Pos=NE.xy+[0.5,-0.5]; Dy=1;
4 Slider("M",[0.000,-5.2],[4,-5.2]);
5 Slider("N",[0.000,-5.7],[4,-5.7]);
6 Text2.xy=[1,-7];
7 mv=M.x/4; nv=N.x/4;
8 Subsedit(2,"mv="+Sprintf(mv,2)
9           +",nv="+Sprintf(nv,2));
10 if(contains(Ch,1),Execstep1());
11 if(contains(Ch,2),Execstep2());
12 if(contains(Ch,3),Execstep3());
13 Windispg();
  
```

Ketjava 20181125
 ketcindylibbasic1[20230813] loaded
 ketcindylibbasic2[20260215] loaded

figures slot

Solution by KeTCindy 2

(3) Description in mkcmd 2

```

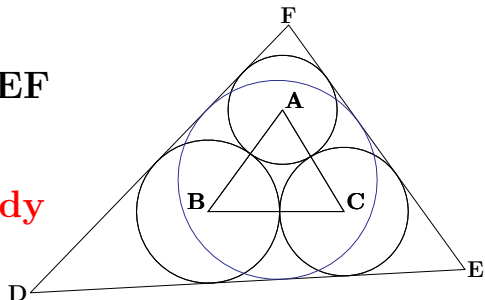
Disptex(Pos,Dy,"ABCをおく",opsb);↓
Disptex([Pos_1+0.5,Pos_2],Dy*0.9,"A::B::C",opsf);↓
Disptex([Pos_1-0.5,Pos_2],Dy,"3円が接する連立方程式",opsb);↓
Disptex([Pos_1+0.5,Pos_2],Dy*0.9,"eq1::eq2::eq3",opsf);↓
Disptex([Pos_1-0.5,Pos_2],Dy,"連立方程式の解",opsb);↓
Disptex([Pos_1+0.5,Pos_2],Dy*0.9,"sol",opsf);↓
Disptex([Pos_1-0.5,Pos_2],Dy,"3円の半径",opsb);↓
Disptex([Pos_1+0.5,Pos_2],Dy*0.9,"r1::r2::r3",opsf);↓
);↓
Dispfig1():=(↓
  Assignvalue();↓
  Parsevv("A::B::C");↓
  Listplot("1",[A,B,C,A],["do"]);↓
  Circledata("1a",[A,r1]);Circledata("1b",[B,r2]);↓
  Circledata("1c",[C,r3]);↓
  Letter([A,"ne","A",B,"nw","B",C,"ne","C"]);↓
);↓

```

Solution from Inside (Strategy)

- (1) Place ABC using putT(m,n,r)
- (2) Create 3 mutually tangent circles C1,C2,C3 centered at A,B,C
- (3) Create common outer tangents using comTan
- (4) Find intersections D,E,F of common outer tangents
- (5) Create incircle of DEF
- (6) Find the inradius

Demo malfattiin.cdy



Solution from Inside (Result)

マウスをドラッグして自由要素を動かす

Figure Parent Ketjson Ketjsoff ParaF Anime Flip

0 1 2 3 4 5 6 7 8 9

NE

m, n, r, iR を r_1, r_2, r_3 で表す

$$\text{sol:} \left[m = \sqrt{\frac{r_1 r_3}{r_2 r_3 + r_2^2 + r_1 r_2}}, n = r_1 / ((r_3 + r_2 + r_1) \sqrt{\frac{r_1 r_3}{r_2 r_3 + r_2^2 + r_1 r_2}}), r = \frac{r_1 r_3}{(r_3 + r_2 + r_1)} \right]$$

$$m: \frac{\sqrt{r_1} \sqrt{r_3}}{\sqrt{r_2} \sqrt{r_3 + r_2 + r_1}}$$

$$n: \frac{\sqrt{r_1} \sqrt{r_2}}{\sqrt{r_3} \sqrt{r_3 + r_2 + r_1}}$$

$$r: (\sqrt{r_1} \sqrt{r_2} \sqrt{r_3}) / \sqrt{r_3 + r_2 + r_1}$$

$$iR: \sqrt{2}((2r_1 r_2^2 + 4r_1^2 r_2 + 2r_1^3) r_3^6 + (7r_1 r_2^3 + 12r_1^2 r_2^2 + 3r_1^3 r_2 - 2r_1^4) r_3^5 + (7r_1^2 r_2^3 + 12r_1^3 r_2^2 + 3r_1^4 r_2 - 2r_1^5) r_3^4 + \dots)$$

$t_1 = \sqrt{r_1}, t_2 = \sqrt{r_2}, t_3 = \sqrt{r_3}$ と置き換え

$$iRs: \frac{t_1 t_2 t_3^3 + \sqrt{t_3^2 + t_2^2 + t_1^2} (t_1 t_2 t_3^2 + t_1 t_2^2 t_3) + t_1^2 t_2 t_3^2 + (t_1 t_2^3 + t_1^2 t_2^2) t_3}{t_2 t_3 \sqrt{t_3^2 + t_2^2 + t_1^2} + t_1 t_3^2 + t_1 t_2 t_3 + t_1 t_2^2}$$

$$iRs2: \frac{t_1 t_2 t_3 (t_3^2 + t_1 t_3 + t_3 + t_2^2 + t_1 t_2 + t_2)}{t_1 t_3^2 + t_1 t_2 t_3 + t_2 t_3 + t_1 t_2^2}$$

$t = \sqrt{t_1^2 + t_2^2 + t_3^2}$ とおき分子を有理化する

$$niR: t_1 t_2 t_3^3 + t_1^2 t_2 t_3^2 + t_1 t_2 t_3^2 + t_1 t_2^3 t_3 + t_1^2 t_2^2 t_3 + t_1 t_2^2 t_3$$

$$niR2: -(2t_1^2 t_2^2 t_3^2 (t_2 t_3 - t_1 t_3 - t_1 t_2) (t_3^2 + t_2^2))$$

$$diR2: -(t_1 t_2 t_3 (t_3 + t_2 + t_1 - t) (t_2 t_3 - t_1 t_3 - t_1 t_2) (t_3^2 + t_2^2))$$

$$iR2: \frac{2t_1 t_2 t_3}{t_3^2 + t_2^2 + t_1^2 - t}$$

$$mv=0.51, nv=0.56$$

Solution from Inside (Discussion)

- $$r = \frac{2\sqrt{r_1 r_2 r_3}}{\sqrt{r_1} + \sqrt{r_2} + \sqrt{r_3} - \sqrt{r_1 + r_2 + r_3}}$$
- Different from the solution in Seiyō Sanpō (1781)

$$2r = \frac{2r_1 \times 2\sqrt{r_2 r_3}}{2\sqrt{r_1(r_1 + r_2 + r_3)} - 2r_1} = \frac{2r_1 \sqrt{r_2 r_3}}{\sqrt{r_1(r_1 + r_2 + r_3)} - r_1}$$
- Also different from the solution in Sanpō Tenzan Tebikigusa (1841)
- Unknown if the error was known early, debated, or corrected later
- Myojorinji Sangaku (1865) correct formula
- Sacred Mathematics (2008) also has the correct formula

Myojorinji Sangaku

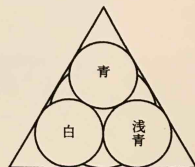
Q As shown, there are 3 circles and a Green circle (incircle) inside the triangle. Given the diameters of the 3 circles, what is the calculation method to find the diameter of the Green circle?

A Find the square root of the sum of the diameters of the 3 circles, call it 'Ten' (Heaven). Subtract 'Ten' from the sum of the square roots of the diameters of the 3 circles, call the remainder 'Jin' (Human). Multiply the diameters of the 3 circles, take the square root, multiply by 2, and divide by 'Jin' to get the diameter of the Green circle.

• Let the diameters of the 3 circles and incircle be d_1, d_2, d_3, D

$$Ten = \sqrt{d_1 + d_2 + d_3}, \quad Jin = \sqrt{d_1} + \sqrt{d_2} + \sqrt{d_3} - Ten$$

$$D = \frac{2\sqrt{d_1 d_2 d_3}}{Jin} = \frac{2\sqrt{d_1 d_2 d_3}}{\sqrt{d_1} + \sqrt{d_2} + \sqrt{d_3} - \sqrt{d_1 + d_2 + d_3}}$$



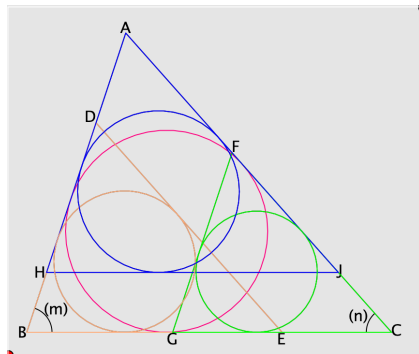
中嶋理光敬愛 志知何某貴忠 謹考
外野郷

今有如図三斜内容緑円及青・白
浅青円、只云浅青径若干白径若干
青径若干問得緑径術如何
答曰、術如左、術曰、置浅青
白青円径和開平方名天浅青白
青径各開平方得商和内減天余
名人置浅青径乘白径及青径開
平方倍之以人除之得緑径合問

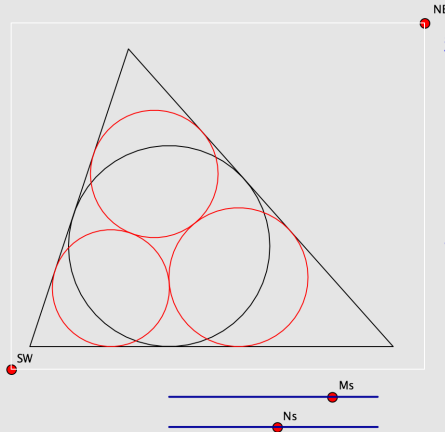
Solution from Outside (Strategy)

- (1) Place ABC using $\text{putT}(m,n,r)$, and let the in-radius be r
- (2) Let r_1, r_2, r_3 be radii of incircles C_1, C_2, C_3 of AHJ, DBE, FGC similar to ABC
- (3) Solve the system of equations derived from mutually tangent conditions of C_1, C_2, C_3

Demo [malfattiout.cdy](#)



Solution from Outside (Result)



$$m=0.72, n=0.45$$

和算家の解答を導く

$$r1: \frac{(N+1)(MN-1)r}{(M+1)(MN-N-M-1)}$$

$$r2: \frac{(M+1)(MN-1)r}{(N+1)(MN-N-M-1)}$$

$$r3: \frac{(M+1)(N+1)(MN-N-M-1)r}{4(MN-1)}$$

置き換え $t_1 = \sqrt{s_1}, t_2 = \sqrt{s_2}, t_3 = \sqrt{s_3},$

$$\text{sol3:} [[M=(2t_1t_2t_3-r)/r, N=(2t_1t_3-r)/r]]$$

$$\text{eq5: } t_1(2t_1^2t_2^2t_3^2-2rt_1t_2t_3^2-2rt_1t_2^2t_3-2rt_1^2t_2t_3+r^2t_2t_3+$$

$$\text{sol5: } [r = -\frac{t_1t_2t_3\sqrt{t_3^2+t_2^2+t_1^2}-t_1t_2t_3^2+(-(t_1t_2^2)-t_1^2t_2)t_3}{(t_2+t_1)t_3+t_1t_2}]$$

$$\text{sol: } r = \frac{t_1t_2t_3\sqrt{t_3^2+t_2^2+t_1^2}+t_1t_2t_3^2+(t_1t_2^2+t_1^2t_2)t_3}{(t_2+t_1)t_3+t_1t_2}$$

$$r: \frac{t_1t_2t_3(\sqrt{t_3^2+t_2^2+t_1^2}+t_3+t_2+t_1)}{t_2t_3+t_1t_3+t_1t_2}$$

Solution from Outside (Discussion)

- Matches the solution of Myojorinji

$$r: \frac{t_1 t_2 t_3 (\sqrt{t_3^2 + t_2^2 + t_1^2} + t_3 + t_2 + t_1)}{t_2 t_3 + t_1 t_3 + t_1 t_2}$$

- Quarter angles play an important role

$$r_1: \frac{(N+1)(MN-1)r}{(M+1)(MN-N-M-1)}$$

$$r_2: \frac{(M+1)(MN-1)r}{(N+1)(MN-N-M-1)}$$

$$r_3: \frac{(M+1)(N+1)(MN-N-M-1)r}{4(MN-1)}$$

Seitz's Derivation Formula

Subtracting x times (6) from (8), and y times (7) from (9), we have

$$b[x \tan \frac{1}{2}B + 2\sqrt{(xy)} + y \cot \frac{1}{2}B] = c[x \tan \frac{1}{2}C + 2\sqrt{(xz)} + z \cot \frac{1}{2}C] \dots\dots\dots (10),$$

$$a[x \cot \frac{1}{2}A + 2\sqrt{(xy)} + y \tan \frac{1}{2}A] = c[y \tan \frac{1}{2}C + 2\sqrt{(yz)} + z \cot \frac{1}{2}C] \dots\dots\dots (11).$$

Multiplying (10) by (4), and (11) by (5), and extracting the square root, we have

$$\sqrt{x} \sin \frac{1}{2}B + \sqrt{y} \cos \frac{1}{2}B = \sqrt{x} \sin \frac{1}{2}C + \sqrt{z} \cos \frac{1}{2}C \dots\dots\dots (12),$$

$$\sqrt{x} \cos \frac{1}{2}A + \sqrt{y} \sin \frac{1}{2}A = \sqrt{y} \sin \frac{1}{2}C + \sqrt{z} \cos \frac{1}{2}C \dots\dots\dots (13).$$

Subtracting (13) from (12), we find

$$\frac{\sqrt{y}}{\sqrt{x}} = \frac{\sin \frac{1}{2}C + \cos \frac{1}{2}A - \sin \frac{1}{2}B}{\sin \frac{1}{2}C - \sin \frac{1}{2}A + \cos \frac{1}{2}B} = \frac{\cos \frac{1}{2}C + \cos \frac{1}{2}(2B+C)}{\cos \frac{1}{2}C + \cos \frac{1}{2}(2A+C)} = \frac{\cos \frac{1}{2}B \cos \frac{1}{2}(\pi-A)}{\cos \frac{1}{2}A \cos \frac{1}{2}(\pi-B)} = \frac{1 + \tan \frac{1}{2}A}{1 + \tan \frac{1}{2}B} \dots\dots (14).$$

$$\begin{aligned} x &= \frac{2c \tan \frac{1}{2}A \tan \frac{1}{2}B (1 + \tan \frac{1}{2}B)}{(1 + \tan \frac{1}{2}A)(\tan \frac{1}{2}A + \tan \frac{1}{2}B)(1 - \tan \frac{1}{2}A \tan \frac{1}{2}B)[1 + \tan \frac{1}{2}(A+B)]} \\ &= \frac{c \sin \frac{1}{2}A \sin \frac{1}{2}B (1 + \tan \frac{1}{2}B)}{\sin \frac{1}{2}(A+B)(1 + \tan \frac{1}{2}A)[1 + \tan \frac{1}{2}(\pi-C)]} = \frac{r(1 + \tan \frac{1}{2}B)}{(1 + \tan \frac{1}{2}A)[1 + \tan \frac{1}{2}(\pi-C)]} \\ &= \frac{\frac{1}{2}r(1 + \tan \frac{1}{2}B)(1 + \tan \frac{1}{2}C)}{1 + \tan \frac{1}{2}A}. \end{aligned}$$

the value of x in (14) and (15), we find

$$y = \frac{\frac{1}{2}r(1 + \tan \frac{1}{2}A)(1 + \tan \frac{1}{2}C)}{1 + \tan \frac{1}{2}B}, \quad \text{and} \quad z = \frac{\frac{1}{2}r(1 + \tan \frac{1}{2}A)(1 + \tan \frac{1}{2}B)}{1 + \tan \frac{1}{2}C}.$$

An excellent solution was furnished by *William Hoover*, and a very brief one by *Professor Casey*. We hope to publish other solutions in a future No.

This celebrated problem has a history, and is known to mathematicians as "Malfatti's Problem."

Equivalence with Seitz's Formula

- Let $L = \tan(A/4)$

$$r1: \frac{(N+1)(MN-1)r}{(M+1)(MN-N-M-1)}$$

$$L = \tan(\pi/4 - B/4 - C/4) = \frac{1 - \tan(B/4 + C/4)}{1 + \tan(B/4 + C/4)}$$

$$= \frac{1 - \frac{M+N}{1-MN}}{1 + \frac{M+N}{1-MN}} = \frac{1-MN-M-N}{1-MN+M+N}$$

$$1 + L = 1 + \frac{1-MN-M-N}{1-MN+M+N} = 2 \frac{1-MN}{1-MN+M+N}$$

$$\frac{1+L}{2} = \frac{1-MN}{1-MN+M+N}$$

$$s1 = \frac{r1}{r} = \frac{(N+1)(MN-1)}{(M+1)(MN-N-M-1)} = \frac{(1+N)(1+L)}{2(1+M)}$$

- Therefore

$$s1 = \frac{(1+\tan C/4)(1+\tan A/4)}{2(1+\tan B/4)}$$

$$y = \frac{\frac{1}{2}r(1+\tan \frac{1}{4}A)(1+\tan \frac{1}{4}C)}{1+\tan \frac{1}{4}B}$$

Conclusion

- Future Work

- Expansion and refinement of the mnR method to handle more complex Sangaku problems (e.g., quarter angles)
- Development of manuals and error-checking functions to ensure beginners can reliably perform interactive solving